

Tailoring allocation policies and improving access to pediatric liver transplantation in Italy: Outframing and concluding

To the Editor:

In Europe, living donor (LD) pediatric liver transplantation (PLT) now accounts for more than one-third of PLT cases.¹ Also, recent meta-analyses have shown that LD-PLT offers many advantages, including lower waiting-list mortality and significantly better graft survival,^{2,3} and during the last decade, the increased use of LD was the only single significant factor that contributed to reduced waiting list mortality in the Eurotransplant area ($p = 0.03$).⁴ Last, a recent Italian study suggests that LD-PLT may play an important role in optimizing both access to PLT and overall outcomes, even in countries with high deceased-donor (DD) donation rates.⁵

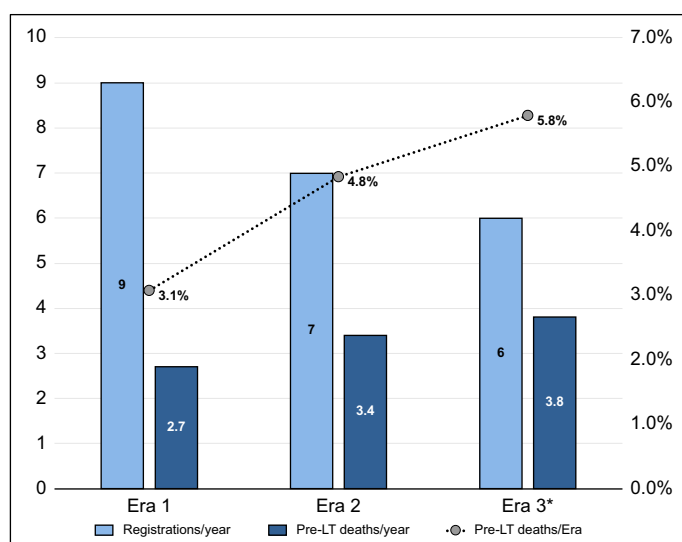
In this other recent analysis, Spada *et al.*⁶ analyzed the dynamics of graft allocation and PLT in Italy during the last two decades. Although 94 LD-PLTs had been performed in that period, they were not considered as a possible contributor to the better outcomes in the recent era. As LD-PLT accounted for up to 15% of all PLT in Era-3 (notably 23.3% in 2018), it clearly contributed to the increased chance of any child receiving PLT and clearly helped to limit the drop-out rate (DOR). Not having taken LD-PLT into account compromises the authors' conclusions.

Era 3 should have been the best era for DD-PLT because i) the "mandatory-split rule" was established in 2015, and ii) it was the era with the lowest number of annual registrations of children who would eventually require DD-PLT ($n = 87, 71$ and 65 in Era-1, Era-2 and Era-3, respectively) (Fig. 1). For the latter group, intention-to-treat survival was 73.5% in fact. This result also speaks against the authors' enthusiasm.

Numbers speak unless the counts are wrong. The legend to Table 3 contains important details that are not considered in the analysis (3/30 patients who were still on the waiting list at December 31, 2021, dropped-out later). This brings DOR in Era-3 to 4.7% in fact, being far above the 2.6% in Era-1, and the 4.3% in Era-2. Moreover, DOR must also be recalculated as candidates who were unregistered because they were too well, and those receiving LD-PLT, were included in the DOR in the authors' analysis. As the latter two patient groups did not really enter the deceased-donor transplant (DD-PLT) allocation process, it would have made more sense to calculate DOR for only those who really needed DD-PLT. After recalculation, this then brings DOR values to 2.7%, 4.8% and 5.6% in Eras-1,-2 and -3, respectively (Fig. 1). This is a concerning observation and, again, is in opposition to the authors' conclusions; the reality is that candidates who are waiting for DD-graft (no LD) face worse outcomes than before.

The national Italian DD-AP has been innovative and very efficient, a fact recognized worldwide. But the authors have missed the point in this analysis; a real alternative hypothesis is that the current overall satisfactory situation is only the direct consequence of rising LD-PLT activity in Italy. This hypothesis may be clarified in the future as a dedicated analysis of LDLT's impact on PLT in Italy is planned in collaboration with the Italian National Transplant Center.

Possibly more concerning is the following evidence from Spada *et al.*'s analysis: i) pre-transplant death occurred mostly in very sick patients, who were however listed as status-3 (an



ALL patients	Era 1	Era 2	Adjusted Era 3*
All registrations	542	560	351
Total LT	515	512	296
LTD	5	40	52**
Total DD-LT	504	472	244**
Unlisted (too well)	12	24	11
Remaining	0	0	1
Pre-LT loss (too sick or dead)	16	24	15

Only patients who required DD-LT liver transplantation	Era 1	Era 2	Adjusted Era 3*
All registrations	520	496	259**
Years in Era	6	7	4
Registration/year	87	71	65
Pre-transplant loss/year	2.7	3.4	3.8
Pre-transplant mortality/Era	3.1%	4.8%	5.8%

Fig. 1. Waiting-list dynamics of children requiring a DDLT in Italy (2002-2018): data from Spada *et al.*⁶ (see "All patients" information). The patients who benefited from LDLT, and those who were de-listed because they were too well were censored. Era 3 was adjusted (*) with data mentioned in the legend of Table 3 by Spada *et al.*⁶ The number of DDLT and LDLT in the adjusted Era 3 (**) was approximated on the basis of the mentioned transplant (TX) ratio LDLT/DDLT in Era 3 (14%). Data are expressed as number or %. DDLT, deceased donor liver transplantation; LDLT, living donor liver transplantation.

elective condition), ii) infants <6 kg have a significantly lower overall chance of success, iii) candidates >25 kg have significantly lower chance of undergoing PLT, and iv) status-2B patients are doing significantly much better than any other group. While such evidence is brought to light, the concern is the absence of any proposal for a revision of the Italian DD-AP. It is probably the major disappointment of the paper. The conclusion could have emphasized that:

- The significantly lower access to PLT for the elective status-3 patients who also carry high DOR (highest rate in the last Era [4.3%]) is a clear call for revising AP as status-3 are supposed to experience the greatest transplant benefit...
- Status-2A patients have a four-fold higher risk of death compared to Status-1. Shouldn't they have priority over status 1? Would this include patients with "acute-on-chronic liver failure"? This has been demonstrated to be the highest-risk group of all.⁷⁻⁹
- The number of "authorized-urgency" cases is rising exponentially (1.1% 3.2%, 7.5% in Eras 1,2,3, respectively), suggesting that the criteria for prioritizing status-3 patients who are deteriorating are too strict.
- Status-2B appears to be overserved compared to status-3 (higher chance of PLT, no drop-out and superior outcome). The status-2B priority needs revision as this group of patients

has grown in an unpredicted manner (11.2% in era 3, $p < 0.001$).

- Children >25 kg are disadvantaged. For these patients, revisiting the full-right-full-left division could be a solution,¹⁰ but the procedure should be prioritized to benefit the >25 kg patients.

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Received 10 January 2024; Received in revised form 15 January 2024;

Accepted 19 January 2024; Available online 13 February 2024

<https://doi.org/10.1016/j.jhep.2024.01.041>

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Financial support

The author did not receive any financial support to produce this manuscript.

Conflict of interest

No conflict of interest – academic opinion work.

Please refer to the accompanying ICMJE disclosure form for further details.

Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhep.2024.01.041>.

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